

第六章 近独立粒子构成的系统.

§ 1

系统:

能级: $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_i, \dots$

简并度: $w_1, w_2, \dots, w_i, \dots$

粒子数: $a_1, a_2, \dots, a_i, \dots$

给出约束: $N = \sum a_i, E = \sum a_i \varepsilon_i$

三种分布的状态数:

1. Boltzmann: $\Omega_{NB} = \frac{N!}{\prod a_i!} \prod w_i^{a_i}$

2. Bose-Einstein: $\Omega_{BE} = \prod \frac{(w_i + a_i - 1)!}{a_i! (w_i - 1)!}$

w_i 选 a_i 个, 可重复. $\rightarrow a_i$ 个星, $w_i - 1$ 个棒.

3. Fermi-Dirac: $\Omega_{FD} = \prod \frac{w_i!}{a_i! (w_i - a_i)!}$

经典极限: $\frac{a_i}{w_i} \ll 1, \frac{\Omega_{NB}}{N!} = \Omega_{BE} = \Omega_{FD}$ (含同性原理)

§ 2 分布的推导.

等几率原理 & 最概然分布. $S = k \ln \Omega$, 为 S 取最大分布.
 $\ln \Omega$ 取最小分布.

Stirling 公式: $\ln(m!) \approx \int_1^m \ln x dx \approx m(\ln m - 1)$

约束: $N = \sum a_i, E = \sum a_i \varepsilon_i$

1. Boltzmann:

$$\ln \Omega = \ln N! + \ln \prod w_i^{a_i} - \ln \prod a_i! \quad (a_i \gg 1)$$

$$\approx N(\ln N - 1) + \sum a_i \ln w_i - \sum \ln a_i!$$

$$\approx N \ln N + \sum a_i (\ln w_i - \ln a_i)$$

$$\delta \ln \Omega = \sum \ln \left(\frac{w_i}{a_i} \right), \delta a_i = 0 \quad (w_i \text{ 不变})$$

考虑 Lagrange 函数:

$$\delta \ln \Omega - \alpha \delta N - \beta \delta E = \sum \left[\ln \left(\frac{w_i}{a_i} \right) - 1 - \beta \epsilon_i \right] \delta a_i = 0$$

$$\Rightarrow \ln \left(\frac{w_i}{a_i} \right) - \alpha - \beta \epsilon_i = 0 \rightarrow a_i = w_i e^{-\alpha - \beta \epsilon_i}$$

$$* \delta^2 \ln \Omega = \sum \frac{a_i}{w_i} \left(1 - \frac{w_i}{a_i} \right) \delta^2 a_i = - \sum \frac{\delta^2 a_i}{a_i} < 0 \quad (a_i > 0)$$

最概然 (实际分布)?

$$\ln \frac{\Omega + \delta \Omega}{\Omega} \approx -\frac{1}{2} \sum \left(\frac{\delta a_i}{a_i} \right)^2 \ll 1$$

2. Bose - Einstein

$$\ln \Omega = \sum \ln (w_i + a_i - 1) - \sum \ln (w_i - 1) - \sum \ln a_i$$

$$\approx \sum (w_i + a_i) \left[\ln (w_i + a_i) - 1 \right] - \sum w_i (\ln w_i - 1) - \sum a_i (\ln a_i - 1)$$

$$= \sum (w_i + a_i) \ln (w_i + a_i) - \sum w_i \ln w_i - \sum a_i \ln a_i$$

$$\delta \ln \Omega = \sum \left[\ln (w_i + a_i) - \ln a_i \right] \delta a_i$$

$$\delta \mathcal{L} = \sum \left[\ln \left(\frac{w_i + a_i}{a_i} \right) - \alpha - \beta \epsilon_i \right] \delta a_i = 0$$

$$\Rightarrow \frac{w_i + a_i}{a_i} = e^{\alpha + \beta \epsilon_i} \rightarrow a_i = \frac{w_i}{e^{\alpha + \beta \epsilon_i} - 1}$$

3. Fermi - Dirac

$$\ln \Omega \approx \sum w_i (\ln w_i - 1) - \sum (w_i - a_i) \left[\ln (w_i - a_i) - 1 \right] - \sum a_i (\ln a_i - 1)$$

$$= \sum w_i \ln w_i - \sum (w_i - a_i) \ln (w_i - a_i) - \sum a_i \ln a_i$$

$$\delta \ln \Omega = \left[\ln (w_i - a_i) - \ln a_i \right] \delta a_i$$

$$\delta \mathcal{L} = \left[\ln \left(\frac{w_i - a_i}{a_i} \right) - \alpha - \beta \epsilon_i \right] \delta a_i = 0$$

$$\Rightarrow \frac{w_i - a_i}{a_i} = e^{\alpha + \beta \epsilon_i} \rightarrow a_i = \frac{w_i}{e^{\alpha + \beta \epsilon_i} + 1}$$

★ 超导混系系统的统计关联效应.

§3 热力学量的统计表达式

1. u, N

$$u = \sum a_i \epsilon_i = \sum \omega_i \epsilon_i e^{-\alpha - \beta \epsilon_i}$$

$$N = \sum a_i = \sum \omega_i e^{-\alpha - \beta \epsilon_i}$$

引入配分函数 $z = \sum \omega_i e^{-\beta \epsilon_i}$

$$\hookrightarrow N = e^{-\alpha} z$$

$$u = -N \frac{\partial}{\partial \beta} \ln z$$

2. w, Q

$$\text{功 } Y = \sum \frac{\partial \epsilon_i}{\partial y} a_i = \sum \frac{\partial \epsilon_i}{\partial y} \omega_i e^{-\alpha - \beta \epsilon_i}$$

$$= \frac{N}{z} \left(-\frac{1}{\beta} \frac{\partial}{\partial y} \right) z = -\frac{N}{\beta} \frac{\partial}{\partial y} \ln z$$

压强 $P = -\frac{N}{\beta} \frac{\partial \ln z}{\partial V} = P(T, V)$, 物态方程.

$$\bar{d}Q = du - \bar{d}w$$

$$= \sum a_i (a_i, \epsilon_i) - dy \cdot \frac{\partial \epsilon_i}{\partial y} a_i$$

$$= \sum \epsilon_i da_i$$

3. 二能级系统.

$$\epsilon_1 = -\epsilon, \quad \epsilon_2 = \epsilon, \quad \omega_1 = \omega_2 = 1$$

对应实例: $S = \frac{1}{2}$ 粒子在磁场中.

$$\text{配分函数 } z = \sum \omega_i e^{-\beta \epsilon_i} = e^{\beta \epsilon} + e^{-\beta \epsilon}$$

$$e^{-\alpha} = \frac{N}{e^{\beta \epsilon} + e^{-\beta \epsilon}}$$

$$\text{平均分布: } \bar{a}_1 = e^{-\alpha + \beta \epsilon} = N \cdot \frac{e^{\beta \epsilon}}{e^{\beta \epsilon} + e^{-\beta \epsilon}}$$

$$\bar{a}_2 = e^{-\alpha - \beta \epsilon} = N \cdot \frac{e^{-\beta \epsilon}}{e^{\beta \epsilon} + e^{-\beta \epsilon}}$$

$$T \rightarrow 0, \quad \bar{a}_1 = N, \quad \bar{a}_2 = 0$$

$$T \rightarrow \infty, \quad \bar{a}_1 = \bar{a}_2 = \frac{N}{2}$$

$$\beta = \frac{1}{kT}$$

$$\bar{E} = \sum a_i \epsilon_i = -N\epsilon \frac{e^{\beta\epsilon} - e^{-\beta\epsilon}}{e^{\beta\epsilon} + e^{-\beta\epsilon}} = -N\epsilon \tanh(\beta\epsilon)$$

$$C = \frac{\partial \bar{E}}{\partial T} = -N\epsilon \frac{1}{\cosh^2(\beta\epsilon)} \cdot \epsilon \frac{\partial \beta}{\partial T}$$

$$= \frac{N\epsilon^2}{kT^2} \frac{1}{\cosh^2(\frac{\epsilon}{kT})}$$

肖特基热容行为, $T \rightarrow 0, C \rightarrow 0$

$$T \rightarrow \infty, C \sim \frac{1}{T^2} \rightarrow 0$$



4. S

$$\alpha S = \frac{1}{T} dQ = \frac{1}{T} (dU - dW) = \frac{1}{T} (dU - \gamma d\eta)$$

$$= \frac{1}{T} \left[d \left(-N \frac{\partial \ln Z}{\partial \beta} \right) - \left(-\frac{N}{\beta} \frac{\partial \ln Z}{\partial \eta} d\eta \right) \right]$$

$$\alpha \ln Z = \alpha \ln Z(\beta, \eta) = \frac{\partial \ln Z}{\partial \beta} d\beta + \frac{\partial \ln Z}{\partial \eta} d\eta$$

$$= \frac{1}{T} \left[-N d \left(\frac{\partial \ln Z}{\partial \beta} \right) + \frac{N}{\beta} \left(d \ln Z - \frac{\partial \ln Z}{\partial \beta} d\beta \right) \right]$$

$$= \frac{N}{T} \left[\frac{1}{\beta} \alpha(\ln Z) - \frac{1}{\beta} d \left(\frac{\partial \ln Z}{\partial \beta} \beta \right) \right]$$

$$= \frac{N}{\beta T} d \left(\ln Z - \frac{\partial \ln Z}{\partial \beta} \beta \right) = k N d \left(\ln Z - \frac{\partial \ln Z}{\partial \beta} \beta \right)$$

$$\Rightarrow S = k N \left(\ln Z - \frac{\partial \ln Z}{\partial \beta} \beta \right) + S'$$

对于 Boltzmann 分布,

$$\frac{\partial \ln Z}{\partial \beta} = -\epsilon$$

$$\ln \Omega \approx N \ln N! + \sum a_i \ln w_i - \sum a_i \ln a_i$$

$$= e^{-\alpha} Z \cdot (-\alpha + \ln Z) - \sum a_i (-\alpha - \beta \epsilon_i)$$

$$= e^{-\alpha} Z \cdot (\ln Z - \alpha) + \alpha e^{-\alpha} Z - \beta \cdot e^{-\alpha} \cdot Z \cdot \frac{\partial \ln Z}{\partial \beta}$$

$$= N (1 - Z - \frac{\partial \ln Z}{\partial \beta} \beta) \quad \frac{\partial Z}{\partial \beta} = -\sum w_i \epsilon_i e^{-\beta \epsilon_i}$$

$$\hookrightarrow S = k \ln \Omega$$

$$= \frac{\partial Z}{\partial \ln Z} \frac{\partial \ln Z}{\partial \beta} = Z \frac{\partial \ln Z}{\partial \beta}$$

5. F .

$$F = U - TS = -N \frac{\partial \ln z}{\partial \beta} - \frac{N}{\beta} (\ln z - \frac{\partial \ln z}{\partial \beta} \beta) \\ = -NkT \ln z$$

6.

在经典极限条件下 $\Omega_{BE} = \Omega_{FD} = \frac{\Omega_{MB}}{N!}$

$$S = k \ln \left(\frac{\Omega_{MB}}{N!} \right) = Nk (\ln z - \beta \frac{\partial \ln z}{\partial \beta}) - k \ln N!$$

$$F = -NkT \ln z + kT \ln N!$$

对二能级系统:

$$T \rightarrow 0 \text{ 时, } a_1 = N, a_2 = 0, \Omega = \frac{N!}{a_1! a_2!} = 1$$

$$S = 0.$$

$$T \rightarrow \infty \text{ 时, } \Omega = \frac{N!}{\left(\frac{N}{2}\right)!^2}, S = k \ln \frac{N!}{\left(\frac{N}{2}\right)!^2}$$

7. 系数 α 的含义

$$\text{化学势 } \mu = \left(\frac{\partial F}{\partial N} \right)_{T, V}$$

$$\text{经典极限下, } F = -NkT \ln z + kT \ln N!$$

$$\approx -kT \left[N \ln z - N (\ln N - 1) \right]$$

$$= -NkT \ln \left(\frac{e z}{N} \right)$$

$$\left(\frac{\partial F}{\partial N} \right)_{T, V} = -kT \left[\ln \left(\frac{e z}{N} \right) + N \cdot \frac{N}{e z} \cdot \left(- \frac{e z}{N^2} \right) \right]$$

$$= kT \ln \left(\frac{N}{z} \right) = -\alpha kT$$

$$\rightarrow \alpha = - \frac{\mu}{kT}$$

8. 理想气体

$$P = \frac{N}{\beta} \frac{\partial \ln z}{\partial V}, \text{ 关键在于求出配分函数 } z$$

对单个粒子,

$$\epsilon = \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2) \quad p_i = \frac{2\pi\hbar}{L} n_i$$

$$n_i \text{ 离散} \rightarrow \text{能级间隔 } \Delta \epsilon \approx \frac{2\pi^2 \hbar^2}{mL^2} \ll 1$$

$$z = \sum_{\epsilon} e^{-\beta \epsilon} = \int_{\text{space}} \frac{dq_x \dots dq_v dp_1 \dots dp_r}{h^3} e^{-\beta \epsilon(p)}$$

对 $r=3$ 的积分:

$$\begin{aligned} Z &= \int \frac{1}{h^3} dx dy dz dp_x dp_y dp_z e^{-\frac{1}{2mkT}(p_x^2 + p_y^2 + p_z^2)} \\ &= \frac{V}{h^3} \left(\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mkT}} dp \right)^3 \quad \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \\ &= \frac{V}{h^3} (2\pi mkT)^{3/2} \end{aligned}$$

或利用高斯积分:

$$\begin{aligned} Z &= \int_0^{\infty} e^{-\frac{\beta}{2m} p^2} \cdot \frac{V}{h^3} 4\pi p^2 dp \\ &= \frac{4\pi V}{h^3} \int_0^{\infty} e^{-\frac{\beta}{2m} p^2} p^2 dp \quad \int_0^{\infty} e^{-\alpha x^2} x^2 dx = \frac{\sqrt{\pi}}{4\alpha^{3/2}} \\ &= \frac{4\pi V}{h^3} \cdot \frac{\sqrt{\pi}}{4 \cdot \left(\frac{\beta}{2m}\right)^{3/2}} = \frac{V}{h^3} (2\pi mkT)^{3/2} \end{aligned}$$

$$\Rightarrow P = \frac{N}{\beta} \cdot \frac{\partial}{\partial V} \ln Z = \frac{N}{\beta} \frac{1}{V} = \frac{NkT}{V}$$

$$PV = NkT$$

$$u = -\frac{N}{Z} \frac{\partial Z}{\partial \beta} = -\frac{N}{Z} \left(-\frac{3}{2}\right) \cdot \frac{Z}{\beta} = \frac{3}{2} NkT$$

$$C_v = \left(\frac{\partial u}{\partial T}\right)_V = \frac{3}{2} Nk$$

$$F = -NkT \ln\left(\frac{eZ}{N}\right)$$

$$= -NkT \left[1 + \frac{3}{2} \ln(2\pi mkT) + \ln\left(\frac{V}{h^3}\right) - \ln N \right]$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V$$

$$= Nk \left[\frac{5}{2} + \frac{3}{2} \ln(2\pi mkT) + \ln\left(\frac{V}{h^3}\right) + \ln\left(\frac{N}{V}\right) \right]$$

9. 经典极限条件: $e^{\alpha} \gg 1$

$$\alpha = -\frac{\mu}{kT} \rightarrow \mu \ll -kT$$

$$e^{\alpha} = \frac{Z}{N} = \frac{V}{N} \cdot \left(\frac{2\pi mkT}{h^2}\right)^{3/2} \gg 1, \quad T \uparrow, V \uparrow, n \downarrow$$

另一种物理解释:

$$\text{热波长 } \lambda_n = \frac{h}{\sqrt{2\pi mkT}} \rightarrow n\lambda_n^3 \ll 1$$

$\Rightarrow$ 粒子无重叠, 非简并.

10. Maxwell 速度分布.

在体积 V 内,

$$N = \int \frac{V}{h^3} dp_x dp_y dp_z \cdot e^{-\alpha - \frac{1}{2mkT}(p_x^2 + p_y^2 + p_z^2)}$$

$$\Rightarrow e^{-\alpha} = \frac{N}{V} \cdot \left(\frac{h^2}{2\pi mkT}\right)^{3/2}$$

$$f(v_x, v_y, v_z) dv_x dv_y dv_z$$

$$= n \cdot \left(\frac{h^2}{2\pi mkT}\right)^{3/2} e^{-\frac{m}{2kT}(v_x^2 + v_y^2 + v_z^2)} dv_x dv_y dv_z$$

$$f(v) dv$$

$$= n \cdot \left(\frac{h^2}{2\pi mkT}\right)^{3/2} e^{-\frac{mv^2}{2kT}} 4\pi v^2 dv$$

$$\frac{df(v)}{dv} \Big|_{v=v_m} = 0 \Rightarrow v_m = \sqrt{\frac{2kT}{m}}$$

$$\bar{v} = \sqrt{\frac{8kT}{\pi m}}$$

$$v_s = \sqrt{\int_0^\infty v^2 f(v) dv} = \sqrt{\frac{3kT}{m}}$$

碰撞次数 Γ

$$d\Gamma \cdot dA dt = f(v_x, v_y, v_z) dv_x dv_y dv_z \cdot v_x dA dt.$$

$$\begin{aligned} \Gamma &= \int f \cdot v_x dv_x dv_y dv_z \\ &= n \cdot \sqrt{\frac{kT}{2\pi m}} = \frac{1}{4} n \bar{v} \end{aligned}$$

11. 能量均分定理.

对于处在温度为 T 的平衡状态的经典系统, 粒子能量中每一个平方项的均值为 $\frac{1}{2} kT$

e.g. 对初解 $\epsilon_p = \frac{1}{2} \sum a_i p_i^2$

$$\begin{aligned}
 \overline{\frac{1}{2} a_j p_j^2} &= \frac{1}{N} \int \frac{1}{2} a_j p_j^2 e^{-\beta \epsilon} \frac{dq_1 \sim dq_r dp_1 \sim dp_r}{h_0^r} \\
 &= \frac{1}{2} \int \frac{1}{2} a_j p_j^2 e^{-\beta \epsilon} \frac{dq_1 \sim dq_r dp_1 \sim dp_r}{h_0^r} \\
 &= \frac{1}{2} \int \frac{1}{2} a_j p_j^2 e^{-\frac{\beta}{2} a_j p_j^2} a_p \cdot \int e^{-\beta \epsilon'} \frac{dq_1 \sim dq_r dp_1 \sim dp_r}{h_0^r} \\
 &= \frac{1}{2} \left(-\frac{1}{2\beta} p_j \cdot e^{-\frac{\beta}{2} a_j p_j^2} \right) \Big|_{-\infty}^{+\infty} + \frac{1}{2\beta} \int e^{-\frac{\beta}{2} a_j p_j^2} a_p \cdot \int e^{-\beta \epsilon'} \frac{dq_1 \sim dq_r dp_1 \sim dp_r}{h_0^r} \\
 &= \frac{1}{2\beta} \int e^{-\beta \epsilon'} \frac{dq_1 \sim dq_r dp_1 \sim dp_r}{h_0^r} \\
 &= \frac{1}{2\beta} \cdot \epsilon = \frac{1}{2} kT
 \end{aligned}$$

应用: ① 单原子分子 忽略了电子对热容量的影响.

$$\epsilon = \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2), \quad \text{因为 } kT \ll eV$$

$$\bar{\epsilon} = \frac{3}{2} kT, \quad U = \frac{3}{2} N kT, \quad C_V = \frac{3}{2} N k, \quad \gamma = \frac{5}{3}$$

② 双原子分子.

$$\begin{aligned}
 \epsilon &= \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2) + \frac{1}{2I} (p_\theta^2 + \frac{1}{\sin^2 \theta} p_\phi^2) \\
 &\quad + \frac{1}{2\mu} p_r^2 + u(r) \quad \text{不考虑振动自由度, 常温不激发.}
 \end{aligned}$$

$$\bar{\epsilon} = \frac{5}{2} kT$$

③ 固体

$$\epsilon = \frac{1}{2m} p^2 + \frac{1}{2} m \omega^2 q^2$$

$$\bar{\epsilon} = 3kT$$

12. 理想气体的内能和热容.

$$\epsilon = \epsilon^t + \epsilon^v + \epsilon^r \quad \text{平动} + \text{振动} + \text{转动}.$$

$$Z = Z^t Z^v Z^r e^{-\beta(\epsilon^t + \epsilon^v + \epsilon^r)}$$

$$= Z^t \cdot Z^v \cdot Z^r$$

$$U = -N \frac{\partial}{\partial \beta} \ln Z = -N \frac{\partial}{\partial \beta} (\ln Z^t + \ln Z^v + \ln Z^r)$$

$$= U^t + U^v + U^r$$

$$C = C^t + C^v + C^r$$

(1) 平动

$$U^t = \frac{3}{2} N k T, \quad C^t = \frac{3}{2} N k$$

(2) 振动 ~ 线性谐振子

$$\epsilon^v = (n + \frac{1}{2}) h \omega$$

$$Z^v = \sum_{n=0}^{\infty} e^{-\beta (n + \frac{1}{2}) h \omega}$$

$$= e^{-\frac{1}{2} \beta h \omega} \frac{1}{1 - e^{-\beta h \omega}}$$

$$U^v = -N \cdot \frac{\partial}{\partial \beta} \ln Z^v = N h \omega \left(\frac{1}{2} + \frac{e^{-\beta h \omega}}{1 - e^{-\beta h \omega}} \right)$$

$$C^v = \left(\frac{\partial U^v}{\partial T} \right)_v = N h \omega \frac{e^{\beta h \omega} - h \omega}{(e^{\beta h \omega} - 1)^2} \cdot \frac{1}{k T^2}$$

$$= N k \left(\frac{h \omega}{k T} \right)^2 \frac{e^{\beta h \omega}}{(e^{\beta h \omega} - 1)^2}$$

引入特征温度: $k \theta_v = h \omega$

$$C^v = N k \left(\frac{\theta_v}{T} \right)^2 \frac{e^{\theta_v/T}}{(e^{\theta_v/T} - 1)^2} \quad \text{一般 } \theta_v \gg T.$$

(3) 转动

$$\epsilon^r = \frac{l(l+1) \hbar^2}{2I}$$

$$Z^r = \sum_{l=0}^{\infty} (2l+1) e^{-\beta \frac{l(l+1) \hbar^2}{2I}}$$

$$k \theta_r = \frac{\hbar^2}{2I}$$

$$= \sum_{l=0}^{\infty} (2l+1) e^{-\frac{\theta_r}{T} l(l+1)}$$

$$\theta_r \ll T$$

$$= \int_0^{\infty} (2l+1) e^{-l(l+1) \frac{\theta_r}{T}} dl$$

$$= \frac{T}{g_r} e^{-\left(1+\frac{g_r}{T}\right)} \Big|_0^\infty = \frac{T}{g_r} = T \cdot \frac{27k}{h^2} = \frac{27}{18h^2}$$

$$u^r = -N \cdot \frac{\partial}{\partial p} \ln \tilde{z}^r = -N \cdot \frac{1}{p} = -Nk]$$